Reg. No. | | | | | | | | | | | [17MT32]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2018

METALLURGICAL THERMODYNAMICS

Time:3 Hours Maximum Marks:70

Instructions:
i. Answer any **FIVE** full Questions choosing atleast two from each part.
ii. Missing data may be suitably assumed.
iii. Use of thermodynamic data hand book is permitted.

PART - A

1. a. Write down the differential equations involving internal energy, enthalpy, Helmholtz free energy and Gibbs free energy for a closed system. Using these differential equations, derive Maxwell's equations for a closed system. 09

b. Define state of a system. Differentiate path function from the state functions. Give examples. 05

2. a. A quantity of supercooled liquid tin is adiabatically contained at 480 K. Calculate the fraction of the tin which spontaneously freezes. 07

Given:

$\Delta Hf = -7070$ J at Tm = 505K

Cp (liquid tin) = $34.7 - (9.2 \times 10^{-3}T)$ J/K

Cp (solid tin) = $18.5 - (26 \times 10^{-3}T)$ J/K

b. Explain the Kirchoff's law in the context of freezing of a pure metal. Determine the latent heat of fusion when one mole of liquid tin freezes at 495K and 485K. 07

Given:

$\Delta Hf = -7070$ J at Tm = 505K

Cp (liquid tin) = $34.7 - (9.2 \times 10^{-3}T)$ J/mol.K

Cp (solid tin) = $18.5 - (26 \times 10^{-3}T)$ J/mol.K

3. a. Write down the differential equation involving G, T and P for a closed system of pure metal. Plot G versus T at constant P and G versus P at constant T for solid and liquid states. Indicate clearly the slopes and phase transformation points on each plot. 06

b. Derive the generalized Gibbs Phase rule for a multi-component system. 08

4. a. Using the following data, calculate the melting temperature of nickel, standard enthalpy of melting, and standard entropy of melting. 07

$NiO(s) = Ni(s) + 0.5O_2(g)$

$\Delta G^0 = -235600 + 86\ T$ Joules (valid between 298-1726 K)

$NiO(s) = Ni(l) + 0.5\ O_2(g)$

$\Delta G^0 = -253090 + 96.1\ T$ Joules (valid between 1726-2200 K)

b. Calculate the vapour pressure of Mg exerted at 1400 °C by the system when the following reaction attains equilibrium. **07**

$4MgO(s) + Si(g) = 2Mg(g) + Mg_2SiO_4(s)$

Given:

$Mg(g) + 0.5O_2(g) = MgO(s)$

$\Delta G^0 = -729600 + 204\,T$ Joules (1363 K $\leq T \leq$ 2200 K)

$2MgO(s) + SiO_2(s) = Mg_2SiO_4(s)$

$\Delta G^0 = -67200 + 4.31\,T$ Joules (298 K $\leq T \leq$ 2171 K)

$Si(g) + O_2(g) = SiO_2(s)$

$\Delta G^0 = -907100 + 175\,T$ Joules (298 K $\leq T \leq$ 1685 K)

PART – B

5. a. Discuss the Roultian, Henriyan and 1 weight percent standard states. Plot the lines for all three standard states. **08**

b. Consider the following reaction at 910°C and 1 atm total pressure. **06**

$[Fe] + H_2O(g) = FeO(s) + H_2(g)$

Calculate the activity of Fe in the alloy, assuming FeO to be pure.

Given:

When Fe is pure, the equilibrium gas composition is 60.2% H_2 and 39.8% H_2O (by volume). If Fe is present as a solid solution of Fe and Ni (XFe = 0.721), the equilibrium gas composition is 59.9% H_2 and 48.1% H_2O.

6. a. At 700K, the activity of Ga in a liquid Ga-Cd solution of composition XGa = 0.5 is 0.79. If the solution exhibits a regular solution behavior, estimate the energy of Ga-Cd bond in the solution. **06**

Given:

i. The molar enthalpies of evaporation of liquid Ga and liquid Cd at their melting temperatures are 270000 and 100000 J respectively.

ii. Co-ordination number of liquid Cd and Ga are 8 and 11 respectively.

b. Derive the expression for molar Gibbs Free energy of mixing of an ideal solution of A and B. Assume both A and B follow Roult's law over an entire compositional range. Using the obtained relationship, show that enthalpy of mixing of such ideal solution is zero. **08**

7. a. Define a Daniel cell. Express the emf of the Daniel cell as a function of activities if the electrolytes are not saturated solutions. Calculate the emf of the cell if the activity of $ZnSO_4$ is twice as large as that of $CuSO_4$. **08**

Given:

$Zn(s) + ZnSO_4(aq) = Cu(s) + CuSO_4(aq)$

$\Delta G^0 = -213040$ J at 25°C.

b. At 1300K, the emf of the following cell is 0.60V. Calculate the activity and activity co-efficient of PbO in the electrolyte. **06**

Pb	$PbO\text{-}SiO_2$	Pt (s), $O_{2\,(g)}$
Pure liquid	Electrolyte $X_{PbO} = 0.5$	$p_{O2} = 1$ atm

Given:

$Pb\,(l) + 0.5O_2\,(g) = (PbO) \quad \Delta G^0 = -88800\ J\ mol^{-1}$

8. a. Give the Avrami equation for kinetics. What is the rate of reaction/transformation according to the Avrami kinetics? If a transformation follows the Avrami kinetics, provide the schematic plot of the following: **07**

   i. Fraction transformed versus the logarithm of time

   ii. Temperature versus the logarithm of $t_{0.5}$

   iii. Temperature versus the rate of transformation

b. State 1st and 2nd order reactions. **07**